Computer Vision for Hand-Strength Prediction in Broadcast Poker

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***Abstract*—Televised poker commentary and a recent broadcast "AI bluff detector" popularized the claim that nonverbal behavior reveals hand strength on camera. We test that claim with a fully automated pipeline over 22 hours of publicly broadcast cash-game footage with exposed hole cards: game state is read from the broadcast graphics by OCR, hands are reassembled into decision windows, hand strength is labeled by Monte Carlo equity, and face and body features are extracted for camera-covered decisions of tracked players. Our central methodological finding is an identity-contamination artifact: under progressively stricter face-attribution hygiene, an apparent behavioral effect first strengthened to a nominally resolvable ΔAUC of $+0.096$ (95% CI $[+0.005, +0.181]$) and then vanished entirely ($+0.009$, $[-0.058, +0.074]$) once every enrolled reference face was verified. The spurious effect was produced by frames in which a lookalike neighbor was silently credited to the tracked player. A preregistered replication on two additional sessions, with the analysis frozen before download, decisively failed: pooled across four sessions, face and pose features change out-of-session AUC by -0.075 (95% CI $[-0.134, -0.020]$, $n = 292$ covered decisions), with approximately 91% power to detect the registered $+0.096$. Sensitivity analyses show the degradation is not an artifact of estimator variance: no regularization scheme rescues the features, a behavior-only model is systematically anti-predictive out of session (AUC 0.349, significantly below chance under a hand-level permutation test, $p = 0.018$), the effect persists within a single player across his four sessions, and only part of the inversion is explained by session-level camera-geometry offsets. Within sessions, by contrast, behavioral signal is marginal and heterogeneous (pooled cross-validated AUC 0.574, 95% CI $[0.492, 0.652]$; per-session 0.42 to 0.68). Whatever tells exist are session-local: carrying them to a new session is measurably worse than ignoring behavior entirely. We conclude that identity hygiene, not feature engineering, is the binding methodological risk for behavioral video datasets built on multi-person footage, and that behavioral hand-strength models of this class do not generalize across sessions.**

***Index Terms*—nonverbal behavior, poker, deception detection, person re-identification, preregistration, broadcast video analysis, negative results**

I. Introduction

Poker folklore holds that players leak the strength of their hands through nonverbal "tells," and the 2026 World Series of Poker broadcast popularized the idea that such leakage is machine-readable, overlaying a behavioral hand-strength model on live coverage. The scientific evidence behind that idea is thinner than its television presence suggests. The best controlled result reports that naive observers judge professional players' hand quality above chance from arm movements alone, at modest effect sizes, while judgments from the face are at or below chance [1]. Automated deception detection benchmarks report within-domain accuracies in the sixties and cross-domain transfer barely above chance [2]. No prior work, to our knowledge, has tested the broadcast-poker claim end to end: does adding automatically extracted nonverbal behavior improve prediction of hand strength over the betting action alone, evaluated out of session, on public footage?

This paper contributes four things.

(1) A reproducible pipeline that converts raw broadcast video into a labeled decision-level dataset: OCR of the broadcast graphics package into per-second game-state snapshots, a state machine that reassembles hands and decision windows, Monte Carlo equity labels from the exposed hole cards, and per-decision behavioral features from face landmarks, blendshapes, and body pose, with per-player normalization and explicit bet-size leakage controls.

(2) An identity-contamination case study (Section V-A). Behavioral datasets built on multi-person footage must attribute every measured face to the right person. We document how three small engineering gaps (an open-set similarity threshold, nondeterministic reference enrollment, and unbounded nearest-position matching) let a lookalike neighbor's face be silently measured as the tracked player's, and how the resulting artifact produced an apparently resolvable behavioral effect that grew under partial cleaning and vanished under full cleaning (Figure 1). We distill the fix into a verification protocol.

(3) A preregistered, adequately powered replication failure (Section V-B). With the analysis frozen before the new footage was downloaded, face and pose features reduce pooled out-of-session AUC by 0.075 (95% CI $[-0.134, -0.020]$). The study had approximately 91% power to detect the registered positive estimate.

(4) A session-locality finding (Section V-C). Behavioral features are not merely uninformative out of session: a behavior-only model scores AUC 0.349, significantly below chance under a hand-level permutation test ($p = 0.018$), and the inversion persists within the same player across his four sessions. Within sessions the same features carry at most marginal, heterogeneous signal (pooled 0.574, CI $[0.492, 0.652]$). Learned behavior-to-strength associations are session-specific in sign: real enough to fit, wrong enough to mislead on a new night. Tells, to the extent our features capture them, are session-local.

II. Related Work

Behavioral cues to poker hand strength. Slepian et al. [1] showed observers rate hand quality above chance from short clips of arm motion (smoothness of the chip push), while face-only clips were directionally below chance. This motivates our smoothness features and our expectation that any true effect is small. Work on automated deception detection in the wild reports within-domain accuracy around 60 to 67 percent on acted and courtroom corpora and severe degradation across domains [2]. Our result extends this pattern to a naturalistic, high-stakes, repeated-game setting with objective ground truth.

Person identification hygiene. Face verification [3] and person re-identification [4] under compression, occlusion (sunglasses are endemic at poker tables), and pose variation are well studied, and deciding whether a detected face belongs to an enrolled identity at all is the classic open-set problem [5]; our initial fixed-threshold gate is exactly the naive open-set design that literature warns against. What is underappreciated is the role of attribution error as a silent noise channel in downstream behavioral datasets: label errors are known to destabilize benchmark conclusions [6], and our artifact is the feature-side analogue, attribution noise manufacturing a positive finding out of another person's face.

Methodological safeguards. We use preregistration [7], grouped bootstrap confidence intervals [8], and leakage review of engineered features. Trajectory smoothness metrics follow [9], [10]; keypoint trajectories are One-Euro filtered [11] before any derivative is taken; face and pose streams come from MediaPipe [12]; OCR uses PP-OCR [13].

III. Data

Footage. Four complete Hustler Casino Live cash-game sessions (February to July 2026, 22 hours total) featuring one high-volume professional (all four sessions) and a second recreational player (one session). The stream exposes hole cards in its graphics, providing ground truth without any inference from behavior.

IV. Methodology

A. Preprocessing and Game-State Extraction

Region-cropped OCR reads the pot, stakes, board, and per-player panels (name, stack, action text, amount to call) once per second; card sprites are template-matched with a saturation-gated segmentation robust to the two background styles in our footage. A state machine folds snapshots into hands and decision windows: a window opens when a player's panel shows an amount to call and closes at their action flip; cutaway gaps are re-merged when positions and hole cards agree; a preroll filter drops inset replays of older sessions. Table I summarizes the result: 722 hands and 5,639 decisions across the four sessions, 4,815 of them with equity labels (Monte Carlo equity versus a random hand, deterministic seeds; six strength classes; `is_weak` marks decisions taken with equity below 0.45, `is_bluff` marks aggressive such decisions).

TABLE I

Dataset summary. A decision is covered when at least half of its window frames pass identity gating (face) or pose tracking for a mapped player; only covered, labeled decisions enter the ablations.

Session	Hours	Snapshots	Hands	Decisions	Labeled	Covered
Feb 2026	5.2	17,501	141	1,320	1,163	114
Apr 2026	5.5	18,814	164	1,188	1,027	55
Jul 2026 (a)	5.2	18,436	225	1,561	1,278	51
Jul 2026 (b)	5.9	20,730	192	1,570	1,347	72
Total	21.8	75,481	722	5,639	4,815	292

B. Behavioral Feature Extraction

For each decision window of a mapped player, frames are searched for the player's face (Section V-A); accepted frames contribute blendshape series (blink rate, smile asymmetry and mean, gaze dispersion, brow activity) and a pose crop that follows the accepted face contributes posture and wrist kinematics (lean variability, head motion, normalized wrist peak speed, spectral arc length, negative log dimensionless jerk, which is amplitude-invariant by construction so it cannot encode bet size). Six event detectors (downward-gaze rate, hand near face, freeze fraction and longest freeze, chip-shuffle periodicity, forward lean) are rates, fractions, ratios, or shoulder-width geometry, and passed the same leakage review; earlier two-session analyses showed them to carry no signal, the preregistration therefore excluded them from the primary test, and Section V-C confirms the null on the pooled data. All features are z-scored per player. Coverage is honest and low: the primary player's face passes identity gating in 10 to 33 percent of window frames depending on session, and 292 labeled decisions meet the 0.5 coverage threshold.

C. Evaluation Protocol

Leave-one-session-out (LOSO): models train on all other sessions and predict the held-out session; predictions are pooled. Baselines use betting features only (position, sizing ratios, pot odds, street, prior raises, stack-to-pot, timing). Deltas are reported with hand-grouped bootstrap 95% CIs (2,000 resamples), resampling hands so correlated decisions within a hand never narrow the interval.

D. Identity Attribution

Attributing faces on a multi-person broadcast requires deciding, per frame, whether a detected face is the tracked player. Our initial design used an open-set rule: accept the first face whose equalized grayscale patch correlates at least 0.55 with any enrolled reference patch of the player, a threshold chosen from a measured same-player/cross-player separation of 0.74 versus 0.25. Section V-A documents how three gaps in this design manufactured a spurious finding; the corrected protocol is: (i) enroll references deterministically, one detector instance

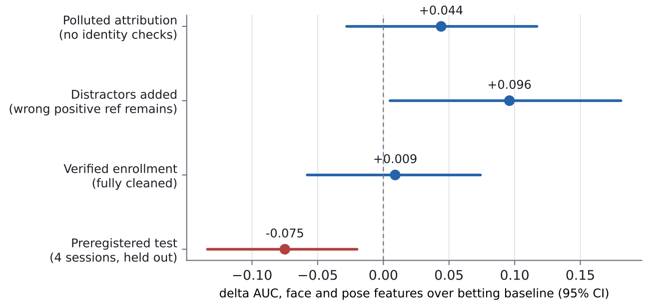

Fig. 1. An apparent behavioral effect tracks attribution pollution, not behavior. Points are ΔAUC of face and pose features over the betting baseline for `is_weak` with hand-grouped bootstrap 95% CIs. The first three estimates are in-sample over two sessions under progressively stricter identity hygiene; the fourth is the preregistered four-session test.

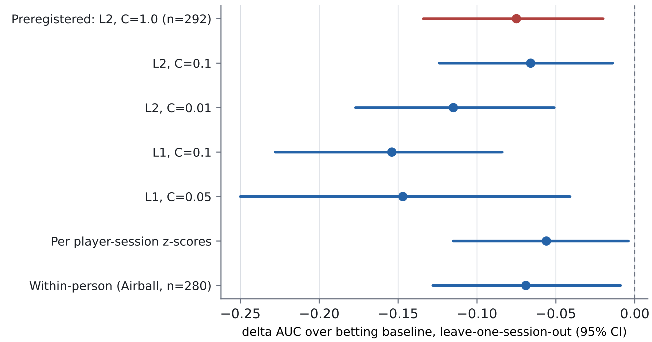

Fig. 2. No analysis choice rescues the features. All arms are ΔAUC over the betting baseline for `is_weak`, pooled LOSO, hand-grouped bootstrap 95% CIs.

per reference frame; (ii) bound enrollment by position and refuse specifications whose intended face is not found; (iii) verify every enrolled patch by eye; (iv) sweep each session for faces that score above threshold against the references, and register every non-target such face as a distractor; (v) attribute a face only if it clears the threshold and beats every distractor score by a margin, best candidate wins; (vi) treat staff and commentators as candidate distractors too (one of ours scored 0.60).

V. Results

A. The Identity-Contamination Artifact

Three gaps in the initial identity design of Section IV-D turned face attribution into a contamination channel.

Gap 1: lookalikes above threshold. A neighboring player correlated 0.66 with the tracked player's references. Whenever the tracked player's face dipped out of detectability (chin on hand, looking down), the neighbor was accepted in his place. In one session, 82 of 193 windows contained some wrong-person frames; one window was entirely the neighbor.

Gap 2: nondeterministic enrollment. Reference patches were enrolled by seeking to hand-verified timestamps and taking the detected face nearest a specified position. Detection under a warm video-mode tracker is stateful: which faces are found at a seek depends on the seeks before it. Re-runs enrolled different faces from the same specification.

Gap 3: unbounded nearest-position matching. When the intended player went undetected at a reference timestamp, the nearest detected face was enrolled with no distance bound. In two of four sessions this placed a neighbor's face inside the player's positive reference library, which also neutralized the distractor defense added after Gap 1: the enrolled lookalike outscored his own distractor reference.

Figure 1 shows the consequence. With no identity checks the apparent effect was $+0.044$ $[-0.028, +0.117]$. After registering the known lookalike as a distractor, but with a wrong face still enrolled as a positive reference, the effect strengthened to $+0.096$ $[+0.005, +0.181]$, nominally resolvable. With enrollment made deterministic (fresh tracker per reference frame), position-bounded (a face must lie within 110 px of its specification), and every enrolled patch visually verified, the in-sample effect collapsed to $+0.009$ $[-0.058, +0.074]$. The mechanism is mundane: another person's face, measured during the tracked player's decisions, covaries with decision context (cameras cut to neighbors during long tanks) in ways that imitate a tell.

B. Preregistered Replication

Before downloading any new footage we froze, in a committed preregistration document: the hypothesis (face and pose features improve pooled LOSO AUC for `is_weak`), the registered estimate ($+0.096$), the exact pipeline settings, the seat-mapping discipline for new sessions (commit-moment identification and a distractor audit before any model output), the pass criterion (pooled CI excluding zero on the positive side), and a prohibition on post-hoc adjustment. Two sessions were selected by roster and layout compatibility; one candidate was excluded before extraction because it used a different graphics package, and the exclusion was logged in the preregistration.

The pooled four-session test returned

$$\Delta\text{AUC} = -0.075, \qquad 95\%\ \text{CI}\ [-0.134,\ -0.020], \qquad n = 292.$$

The CI excludes zero on the negative side: added behavioral features reduce out-of-session performance. Under a normal approximation to the bootstrap distribution (SE ≈ 0.029), the test had approximately 91% power to detect the registered $+0.096$ and 80% power for any delta of at least $+0.082$; the upper confidence bound rules out all positive effects at the 2.5% level. The betting-only baseline itself scores AUC 0.549 pooled out of session, against 0.75 within session, replicating the known brutality of cross-session transfer.

C. Model Results: Why Negative, Not Just Null

A significantly negative delta invites the objection that the estimator, not the signal, is at fault: ten added dimensions can hurt a small model even when they carry information. Four analyses reject this reading (Figure 2).

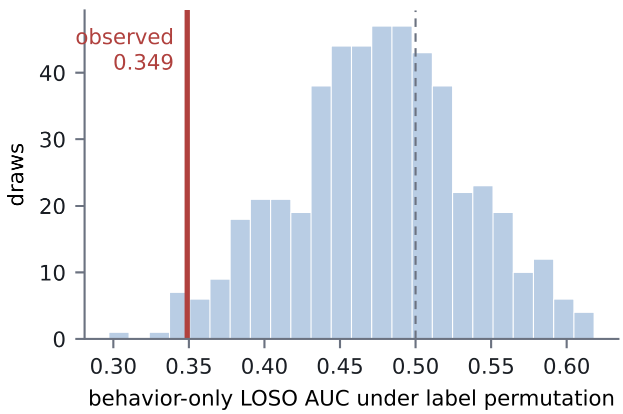

Fig. 3. The below-chance transfer is significant. Hand-level label permutations within each session, re-running the full cross-session pipeline per draw (500 draws), place the observed behavior-only AUC in the null's lower tail ($p = 0.018$).

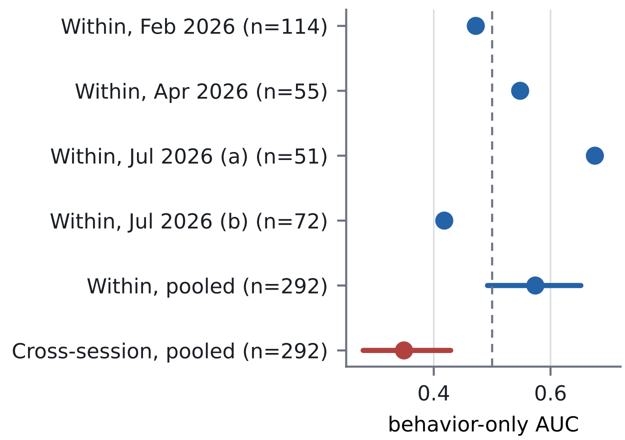

Fig. 4. Session locality. Within-session grouped cross-validation finds marginal, heterogeneous behavioral signal; the same features transfer significantly below chance across sessions.

Regularization does not help. Stronger L2 shrinkage leaves the delta unchanged or worse ($C = 0.01$: -0.115), and sparse L1 selection is worst of all (-0.154): selecting the few behavioral features that look best in training amplifies reliance on associations that do not hold in the held-out session.

Behavior alone is anti-predictive. A behavior-only model scores AUC 0.349, far below chance. Whatever association the model learns between behavior and weakness in three sessions, the reverse tends to hold in the fourth. Under label exchange this would be signal; under session exchange it is instability.

The inversion survives within a person. Restricted to the primary player's four sessions (one identity, one seat style, $n = 280$), the delta is -0.069 $[-0.128, -0.009]$. Idiosyncrasy across people does not explain the failure; instability within a person across nights does.

Session geometry explains only part. Re-standardizing features per player-session (removing session-level offsets induced by camera geometry, since a lean angle or apparent motion scale depends on the shot) moves the behavior-only AUC from 0.349 to 0.387 and the delta to -0.056 $[-0.115, -0.004]$. Session offsets are therefore a real confound channel that future work must normalize away, and still not the whole story.

The inversion is significant, and within-session signal is marginal. Two further analyses locate the failure precisely. First, a permutation test that shuffles labels at the hand level within each session and re-runs the full cross-session pipeline places the observed behavior-only AUC of 0.349 (hand-grouped bootstrap CI $[0.279, 0.429]$, entirely below chance) in the lower tail of its null (null mean 0.476, SD 0.058, $p = 0.018$; Figure 3): transfer is not merely absent but systematically inverted. Second, within-session grouped cross-validation (5-fold by hand, per session, pooled) gives a behavior-only AUC of 0.574 with a hand-grouped bootstrap CI of $[0.492, 0.652]$, with per-session values ranging from 0.418 to 0.676 (Figure 4). Together these say the model is not hallucinating: it finds real within-session structure in at least some sessions, but the sign of that structure is session-specific, so exporting it is worse than ignoring behavior. This is the paper's clearest positive characterization: behavioral hand-strength associations in this setting are session-local.

The event detectors are a true null, not an inversion. As secondary arms on the pooled four-session data, the six leakage-proofed event detectors alone give $+0.011$ $[-0.037, +0.063]$, and the union of all behavioral features gives -0.060 $[-0.128, +0.005]$. The contrast with face and pose is instructive: the low-dimensional detectors, built as rates and ratios with explicit leakage review, neither help nor hurt, while the richer face and pose summaries are the source of the inversion.

VI. Discussion

Three readings of these results deserve separation. The narrow reading is engineering: with tabular summary features over MediaPipe streams, a linear model, four sessions, and camera-limited coverage, behavioral signal is not extractable, and the attempt costs accuracy. The middle reading is statistical: effects of the size the literature supports (a few hundredths of AUC) require an order of magnitude more covered decisions than 22 hours of broadcast yields, and any study in this regime that reports a large positive behavioral effect should be audited first for identity contamination, which we show can manufacture exactly such an effect. The broad reading is substantive: even within one professional, behavior-strength associations invert across nights, which is what one expects if experienced players' surface behavior is dominated by context, table dynamics, and deliberate randomization rather than stable leakage. Our data cannot distinguish "no tell exists" from "tells exist but are contextual and unstable"; both entail that broadcast-scale models of this class cannot use them.

The artifact deserves the emphasis we give it. It produced, in our own hands, a preregistration-worthy positive estimate with a confidence interval excluding zero, out of nothing but face misattribution, and it grew stronger under partial cleaning because the partially cleaned pipeline retained the most correlated contamination. Behavioral-video findings built on multi-person footage without an explicit attribution audit should be treated as provisional.

VII. Limitations

One primary player carries most covered decisions; sunglasses make eye features noise for him, as for many poker subjects. Coverage is bound by broadcast camera direction, not by us. Card-reading errors survive at a low rate (rank confusions on small sprites) and add label noise to equity. The betting baseline is deliberately simple; a stronger baseline would only shrink headroom for behavior. Our features are window-level summaries; sequence models over raw streams remain untested here, though they face the same coverage and identity constraints with more capacity to overfit them.

VIII. Conclusion

We built an end-to-end pipeline from broadcast poker video to labeled decision-level behavioral data and asked whether nonverbal behavior adds predictive power over betting action for hand strength. The answer, under a preregistered and adequately powered test, is that it subtracts: behavior-strength associations are learnable within a session but their signs are session-specific, so models that carry them to a new session perform significantly below chance. Along the way we showed how easily the opposite conclusion is manufactured: a single misattributed face in a reference library produced a nominally significant positive effect that survived partial cleaning. The attribution-hygiene protocol, the preregistration, and the full analysis history are released with the code. For behavioral video research, the order of operations this study recommends is: verify identity first, preregister second, and only then believe an effect.

IX. Ethics and Reproducibility

All footage is publicly broadcast with player consent to the stream's hole-card exposure; we analyze it post hoc, redistribute no footage or identifiable imagery, and publish code, derived features, timestamps, and the preregistration rather than clips. Pipeline, analysis code, seat maps, audit tooling, figure scripts, and the frozen preregistration with its logged result are available at https://github.com/minuuva/Poker-Bluff-Detector (private during review, public with publication); the full analysis history, including the artifact's discovery and every estimate in Figure 1, is preserved in that repository's commit log.

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